GARDEN SPRINKLER CONTROLLER

Developed by Wally Horsman, 2026

Seven garden zones and an automatic channel 8 master pump, using the Waveshare Modbus POE ETH Relay (B).

BACKGROUND OPERATION

The controller now runs as a Linux background service. It starts at boot, continues while logged out, and does not need the desktop window open. Open the desktop app whenever you want to see status or change settings. Closing the window does NOT stop watering. Use RAIN PAUSE or STOP ALL. Only the service communicates with the relay board.

Requirements: Linux with systemd and Python 3.11 or newer. The desktop also needs Tk and CustomTkinter. The service needs no graphical display.

INSTALLATION

Close any older direct-hardware version of the sprinkler app first. The supplied unit expects the project at:

```
~/Documents/garden-sprinkler-controller
```

From the project folder:

```
python3 -m venv .venv
.venv/bin/python -m pip install -r requirements.txt
mkdir -p ~/.config/systemd/user
cp garden-sprinkler.service ~/.config/systemd/user/
loginctl enable-linger "$USER"
systemctl --user daemon-reload
systemctl --user enable --now garden-sprinkler.service
```

Enabling linger may need administrator approval on another machine. It allows your service manager to start before login and survive logout. Do not move/delete the project or virtual environment while installed.

Open the desktop:

```
.venv/bin/python app.py
```

The desktop connects to a private same-user local socket. There is no network control listener and no direct-hardware fallback. Opening/reopening the desktop does not reset or interrupt the relays.

Check status:

```
systemctl --user status garden-sprinkler.service
```

Read recent service errors:

```
journalctl --user -u garden-sprinkler.service -n 50 --no-pager
```

Restart after checking and correcting a hardware fault:

```
systemctl --user restart garden-sprinkler.service
```

Disable automatic startup and stop the service:

```
  systemctl --user disable --now garden-sprinkler.service

POWER FAILURE AND RECOVERY
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The enabled service starts when the machine boots, even before login.
Failed startup, such as waiting for the network or relay box, retries
every 15 seconds. A runtime hardware fault stops scheduling, attempts
all outputs OFF, logs the fault, and requires a service restart.
It does not repeatedly retry ON commands after a runtime fault.

At service startup all eight outputs are commanded OFF, pump first.
Active saved schedules then resume for their remaining time unless
Rain Pause is enabled. Missed watering windows are not replayed.
Interrupted manual/test runs do not restart.

The computer must have power, network access, and remain awake.
This installation does not change system sleep or firmware settings.
If necessary, enable "Restore on AC Power Loss" or "AC Back" in the
computer's BIOS/UEFI so it powers up when mains power returns.
Software cannot switch on a powered-off computer.

SETTINGS
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Default schedule file:

  ~/.config/garden-sprinkler-controller/settings.json

Rain Pause is saved separately beside it:

  settings.json.rain-pause.json

Include BOTH files in backups. Missing schedules start disabled.
A missing pause file means not paused. Invalid files produce errors;
they are not silently overwritten. Writes use atomic replacement.
Older version 1, 2 and 3 schedules are preserved and saved as version 4.

The service owns settings writes. To change configuration paths or
relay parameters, use:

  systemctl --user edit garden-sprinkler.service

Override ExecStart, clearing its previous value first with ExecStart=.
The --config, --host, --port, --slave-id and --timeout arguments belong
to service.py, not app.py. Restart the service after changes.
For foreground diagnosis, stop the installed service before running:

  .venv/bin/python service.py

The desktop accepts --socket for an alternate local service endpoint.

USING THE DESKTOP
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Daily Schedule, above the editor, opens a weekday table of all
configured garden periods sorted by start time. It includes unsaved
edits, clearly labelled, and excludes the master pump. Disabled routines
are marked and can be hidden. Select a row to open that relay's settings.
Amber warnings flag identical starts and overlapping periods across
enabled relays. Adjacent end/start times do not overlap. Warnings are
advisory and do not block saving: when two enabled, unpaused schedules
overlap, both relays can run at the same time. Watering is not queued
sequentially; the shared master pump runs for their combined demand.
```

Invalid/incomplete pairs remain visible with a warning; they cannot be fully checked for conflicts. Rain Pause and stale service status are labelled. The table shows configured times, not current relay outputs.

Weekly Overview, beside Daily Schedule, shows all garden relays and all seven weekdays at once. Each cell lists four numbered start-stop pairs; unused periods show --. Bright red text and ! flag same-start or overlapping periods on both affected relays; red cells flag invalid/incomplete pairs. Click a garden or cell to open its settings.
Both tables can stay open together and update as you edit. Both can hide disabled routines and label unsaved edits. Conflicts compare the same weekday across enabled garden relays, not different days.

The compact opening window is 940 x 760, with a minimum size of 940 x 620. One RUN ZONE / STOP THIS ZONE button handles timed watering and stopping. STOP ALL and Save Settings stay fixed at the bottom. At smaller heights, scroll the configuration panel.
Watering controls and Rain Pause remain outside the scrolling panel.

Select a garden zone (1-7), edit its Relay / Garden Name, and save. Each name can have up to 40 characters.

Each weekday has four independent start/stop pairs, shown in muted blue, green, amber and purple. Enter 24-hour times as HMM (815), HHMM (0815), or HH:MM (08:15). Times display as HH:MM.
Leave both fields blank to skip a window. Every used pair needs both times. Stop must be later than start. Enter windows in any numbered order, but do not overlap windows within one zone; windows cannot cross midnight. Click a Start heading to pause/resume that window number for the selected zone across all weekdays. The smaller Paused marker appears below paused headings. Changes apply on save. Red borders identify overlap fields. Clicking a conflicting Stop field opens an acknowledgment popup naming both groups.

"Copy selected day's times to all days" copies all four pairs once. Every day remains independently editable afterward.

Save Settings applies all garden zones and preserves the unused channel 8 settings. Unsaved edits survive zone changes but do not alter watering. Closing prompts to save/discard edits, not to stop the service. Other desktop windows receive saved changes when they have no unsaved edits. Stale saves are rejected; close/reopen that window before editing again to avoid overwriting another window's changes.

Status dots:

Red	Relay commanded/confirmed ON.
Bright green	Current time is in an enabled saved schedule, but relay is OFF (for example, its current window was stopped).
Grey	No active schedule, disabled routine, or Rain Pause.
Fault colour	Status unknown; check the service and device.

RUN ZONE waters the selected zone for up to 10 minutes, even if its routine is disabled. Its button becomes STOP THIS ZONE while watering.

STOP THIS ZONE and STOP ALL suppress currently active schedule windows until they end. Later windows still run. Suppression survives closing the desktop, but not restarting the service/computer. Manual runs can still be started unless Rain Pause is enabled.

RAIN PAUSE

RAIN PAUSE immediately sends all eight outputs OFF and prevents scheduled, manual, and test watering. It saves automatically without

saving or discarding schedule edits, and survives service/PC restarts.
Schedules remain unchanged. Turning pause OFF asks for confirmation;
an active saved schedule may start immediately for its remaining time.
Previous stop suppression still applies. Manual/test runs do not resume.

If the app loses its service connection, status becomes UNKNOWN and
watering controls are disabled. It retries the local connection every
second. The service may still be watering; do not assume outputs are OFF.
A timed-out command is not automatically replayed because it may already
have succeeded. Status polling reconciles the result.

CHANNEL 8 MASTER PUMP

Channel 8 is labelled Master Pump, automatically ON whenever any garden
zone (1-7) is watering, including manual/test operation. It has no
independent schedule or watering controls.

The first valve opens before pump ON. Pump stays ON while any zone runs,
and stops before the last valve closes. Startup, STOP ALL, Rain Pause,
fault shutdown and normal service shutdown attempt pump OFF first,
then all seven valves. There are no valve-travel or pump delays.
Command order does not prove a physical valve has finished moving.

The old channel 8 name and schedule are preserved but ignored.
To restore eight independent garden zones:

1. Stop the service.
2. In controller.py, set MASTER_PUMP_CHANNEL = None.
3. Restart the service and reopen the desktop.

The nearby code comment explains this setting. The original scheduling
and editing logic remains available. Review the old channel 8 routine
before restoring it; an enabled active routine can start immediately.

HARDWARE AND SAFETY

Confirmed device parameters:

IP address	192.168.0.99
TCP port	4196
Work mode	TCP Server
Transfer mode	None (Modbus RTU-over-TCP, NOT Modbus TCP framing)
Slave ID	1
Internal serial	115200 baud, 8 data bits, no parity, 1 stop bit

Zones 1-7 use coils 0-6. Master pump uses coil 7.
The adapter checks CRC, response address/function, write echoes and
coil readback. ON commands are never automatically retried.
Readback confirms logical relay state, NOT valve movement or water flow.
Do not use competing controllers or incompatible board linkage modes.

Optional read-only checks (no writes):

```
python3 waveshare_relay.py
python3 waveshare_relay.py --discover
```

Discovery tries slave IDs 1-247 and stops at the first valid response.
It may take several minutes; cancel with Ctrl+C.

Scheduling uses the computer's local time/day, checked every 250 ms
when idle. Network/IPC operations can delay checks; this is not real
time. Manual/test durations use a monotonic clock.

Clock adjustments and daylight-saving changes can shorten/skip runs.
Multiple zones may water together; there is no sequential queue or
flow/pressure monitoring.

Normal service shutdown sends all outputs OFF. A killed service,
network failure, sleeping PC or frozen process cannot guarantee OFF.
Timers are SOFTWARE ONLY, not a hardware watchdog.
Use normally-open contacts and an independent hardware cutoff.
Check channel mapping and shutdown with valve power disconnected first.
Never rely on this app alone to prevent flooding or protect a pump.
Use suitably rated switching/contactors for pump starting current.
Have mains wiring and protection checked by a qualified electrician.
Independent dry-run, pressure and overload protection may be needed.

TESTS

Core, transport and service tests (no display needed):

```
python3 -m unittest -v test_controller test_waveshare_relay test_service
```

All tests, including the desktop (Tk/CustomTkinter and display needed):

```
.venv/bin/python -m unittest -v
```

Tests use fake hardware and local loopback/Unix sockets. They do not
operate the physical relay board.